DIRECT SUMMANDS OF RELATIVELY FINITELY PRESENTED MODULES ARE RELATIVELY FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16557-16567 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 16569-16574, the complete author proof (4 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Translation-protocol template; slogan environment retained in the pinned statement and shipped as-is.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. More on Algebra, lemma sum-relatively-finite-presentation; the relative-finite-presentation dictionary (sibling card L16180).

Source reference. Original lemma 0672: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16574–16574. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A direct summand of a module inherits the property of being finitely presented relative to a base. Let $X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}, \mathcal{F}'$ be quasi-coherent $\mathcal{O}_X$-modules. If $\mathcal{F} \oplus \mathcal{F}'$ is finitely presented relative to S, then so are $\mathcal{F}$ and $\mathcal{F}'$.*

Proof. Translation of the result of More on Algebra, Lemma 0672 into the language of schemes. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.